Agentic Chatbot – Final Features & Technical Roadmap

I've combined the original proposal features with some extended, innovative ideas. The document is organized into three sections: **Core Features** (those achievable within the project scope), **Enhanced Features** (those feasible with moderate effort), and **Future Extensions** (advanced, ambitious concepts). For each feature, I've included the recommended technologies and a practical roadmap for implementation.

Part A: Core Features (Proposal-Aligned)

GPT-Based Q&A Chatbot

Tech Stack

- LangChain (framework for chaining LLM calls)

- OpenAI GPT-4/4o API (core LLM)

- FastAPI/FastMCP server (backend)

Implementation Roadmap

- Setup backend with FastMCP for structured tool calls.

- Integrate LangChain conversational chain with OpenAI API.

- Add memory (ConversationBufferMemory).

- Build FAQ knowledge base + fallback to GPT.

- Integrate into frontend (React, Discord bot, or Teams).

Document Retrieval (RAG)

Tech Stack

- ChromaDB (vector database for embeddings)

- OpenAI embeddings (text-embedding-3 models)

- PyMuPDF / pdfplumber for parsing PDFs

- Google Drive API

Implementation Roadmap

- Use Google Drive API for authentication and file listing.

- Parse text from docs and chunk into 500–1000 tokens.

- Embed chunks with OpenAI API and store in ChromaDB.

- On query: embed input → search ChromaDB → return docs → GPT summarization.

- Add metadata filtering (author, date, tags).

Knowledge Graph (Neo4j)

Tech Stack

- Neo4j (graph database)

- Neo4j Python driver

- LangChain GraphQA

Implementation Roadmap

- Define schema: (Member) – [HAS_SKILL] → (Skill), (Member) – [WORKS_ON] → (Project).

- Populate graph with Robotics Club data.

- Enable queries like 'Who is working on Project X?'.

- Add skill progression mapping and analytics.

Agentic Workflows (LangGraph)

Tech Stack

- LangGraph

- FastMCP

Implementation Roadmap

- Define tools: doc search, Gmail, Google Forms.

- Build simple 2–3 step workflows (e.g., search doc → email).

- Add retry/error handling nodes.

- Optimize by caching results (Redis optional).

Google Services (Gmail, Forms, Calendar)

Tech Stack

- Google Gmail API

- Google Forms API + Sheets API

- Google Calendar API

Implementation Roadmap

- Integrate Gmail API for automated mail notifications.

- Provide pre-filled Google Forms links + sync responses with Sheets.

- Use Calendar API for event reminders and room bookings.

Task Tracking & Skill Evaluation

Tech Stack

- Neo4j

- Pandas

- Matplotlib/Plotly

Implementation Roadmap

- Store tasks in Neo4j as nodes linked to members.

- Log task completion and update skill scores.

- Generate analytics dashboards using Pandas + visualization libraries.

- Expose progress reports via chatbot or web dashboard.

Event Conduct & Evaluation

Tech Stack

- Pandas

- ReportLab/WeasyPrint

- Workflow scripts

Implementation Roadmap

- Define judging templates (criteria JSON).

- Collect scores via Google Forms/web interface.

- Aggregate results using Pandas and generate rankings.

- Auto-generate reports and certificates (PDF).

Part B: Enhanced Features (Feasible Additions)

Real-Time Notifications & Milestones

Tech Stack

- Neo4j for project–task–member graph

- Microsoft Teams API (incoming webhooks)

- LangGraph for notification bar logic

- Celery/cron jobs for scheduling

Implementation Roadmap

- Model tasks, projects, deadlines in Neo4j.

- Send task assignment/deadline alerts via Teams webhook.

- Use Celery/cron jobs for deadline reminders.

- Trigger milestone-achieved notifications when Neo4j updates.

Profile Personalisation

Tech Stack

- Neo4j

- Google Forms API

- Teams API

Implementation Roadmap

- Collect event participation data from Forms.

- Link members to events in Neo4j.

- Fetch Teams IDs via API for member profiles.

- Show contribution stats (events, solo/team tasks).

Project Status Reports

Tech Stack

- Neo4j (task + project graph)

- Google Drive API (project documents)

- GitHub API (commits, repos)

- Gmail API

Implementation Roadmap

- Fetch recent updates from Google Drive.

- Check commits from linked GitHub repos.

- Correlate with tasks in Neo4j to track project health.

- Generate automated mail summary to project members.

Part C: Future Extensions (Ambitious Ideas)

Vision-Based Hardware Debugging

- Google Vision API or GPT-4 Vision for detecting errors in hardware design.

- Enable real-time query handling with image uploads or screen sharing.

Voice Assistant

- Speech recognition (Google Cloud Speech-to-Text, speech_recognition lib).

- Text-to-speech (pyttsx3 or ElevenLabs).

Interactive Code Execution

- FastAPI + WebSocket API for live responses.

- JupyterLab API to run code snippets and return results.

- Teams/Discord notifications for interactive cells.

Query Classification & Routing

- spaCy/NLTK or HuggingFace Transformers for intent classification.

- LangGraph agents for routing queries to the correct tool.

Leaderboard Automation

- Redis-py for fast in-memory score updates.

- FastAPI for API exposure.

- Auto-generate leaderboards during competitions.